

GATI SHAKTI VISHWAVIDYALAYA

C Programming Language

Lab Assignment – 3

Instructions:

1. Write all the program and execute.
 2. Maintain soft copy of all codes and prepare a Lab Record consisting hand written programs (10 Programs out of this list).
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1. Write a program to display fibonacci series where number of terms should be entered by the user.

0 1 1 2 3 5 8 13 21

2. Write a program to print factorial of a number entered by the user.
3. Write a program to read characters until an * is encountered. Also, count the number of uppercase, lowercase, and numbers entered by the user.
4. Write a program to print the following pattern :

A
AB
ABC
ABCD
ABCDE

5. Write a program to print following pattern:

1
121
12321
1234321

6. Write a program to print the sum of the series where number of terms should be entered by the user :

- a) $1/2 + 2/3 + 3/4 + \dots + n/n+1$
- b) $1^2/1 + 2^2/2 + 3^2/3 + \dots$
- c) $1 + (1+2) + (1+2+3) + \dots$

7. Write a program to read and display n numbers using array.
8. Write a program to find smallest of n numbers using array.
9. Write a program to delete an element from an array.
10. Write a program to interchange the largest and smallest number in the array.
11. Write a program to read marks of 10 students in the range of 0-100. Then make 10 groups: 0-10, 11-20, 21-30, so on. Count the number of values that falls in each group and display the result.

12. Write a program to put even & odd elements of an array in 2 separate arrays.
13. Write a program to swap two numbers using bitwise operators.
14. Write a program to implement merging of two sorted arrays.
15. Write a program to copy the contents of one array into another in the reverse order.